Module 0 — Linux Setup & Toolchain

Supplementary to the sPHENIX Standalone Course

1. System Requirements

This course assumes a single Linux workstation with at least 20 GB free disk, 8 GB RAM, and a dual-core CPU. More cores mean faster Geant4 builds (the dominant time cost) but are not required. A laptop is fine.

Requirement	Minimum / Recommended
OS	Ubuntu 22.04 / 24.04 LTS, Debian 12, Fedora 39+
Disk	20 GB free (Geant4 data ~3 GB, builds ~8 GB)
RAM	8 GB minimum; 16 GB comfortable
CPU	Any x86_64; ARM64 also supported by all components
Internet	Needed for downloads; no internet needed after install

2. Package Installation

Debian / Ubuntu

```
sudo apt update
sudo apt install -y build-essential cmake git wget curl \
    gfortran gdb valgrind \
    libx11-dev libxpm-dev libxft-dev libxext-dev \
    libssl-dev libpcre3-dev xlibmesa-glu-dev libglew-dev \
    libftgl-dev libmysqlclient-dev libfftw3-dev libcfitsio-dev \
    graphviz-dev libavahi-compat-libdnssd-dev libldap2-dev \
    python3 python3-dev python3-numpy libxml2-dev libkrb5-dev \
    libgsl0-dev qtwebengine5-dev
```

Fedora / RHEL / AlmaLinux

```
sudo dnf groupinstall -y "Development Tools"
```

```
sudo dnf install -y cmake git wget gcc-gfortran gdb valgrind \
    libX11-devel libXpm-devel libXft-devel libXext-devel openssl-devel \
    mesa-libGLU-devel glew-devel ftgl-devel mariadb-devel \
    fftw-devel cfitsio-devel graphviz-devel gsl-devel \
    python3-devel python3-numpy libxml2-devel krb5-devel
```

3. Known-Good Tool Versions

All known-good combinations verified in April 2026 on Ubuntu 24.04 and Fedora 40:

Tool	Version used in examples
gcc / g++	11.4 — 13.3
gfortran	11.4 — 13.3
cmake	3.22 or newer (3.27 recommended)
ROOT	6.28/04 or 6.30/02
Pythia8	8.312
Geant4	11.2.1 (LTS) or 11.3.0
HepMC	2.06.11 (for Pythia8 writer) + HepMC3 3.2.6
Python	3.10 — 3.12

4. Disk Budget

Plan for about 15 GB of installed software. A typical breakdown:

Component	Approximate on-disk size
ROOT build + install	~1.2 GB
Pythia8 build	~150 MB
HIJING build	~20 MB
Geant4 source + build	~4 GB
Geant4 data files	~3 GB
HepMC2 + HepMC3	~80 MB
Your output data	1–5 GB per production

Note: Use an SSD if you have one — the main bottleneck during the Geant4 build is I/O on small files. An NVMe SSD reduces build time by 30–50% compared to a spinning disk.

5. Directory Layout (recommended)

```
$HOME/
■■■ src/                <- source tarballs, cloned repos
■   ■■■ root/
■   ■■■ pythia8/
■   ■■■ geant4/
■   ■■■ hijing/
■■■ opt/                <- installed prefixes
■   ■■■ root/
■   ■■■ pythia8/
■   ■■■ geant4/
■   ■■■ hepmc/
■■■ work/               <- analysis code, scratch
■■■ data/               <- HepMC files, ROOT files
```

Keep **source**, **opt**, and **work** strictly separated so you can rebuild or remove one tool without breaking the others.

6. Shell Environment

Create ~/.sphenixrc that sources all tool environments:

```
#!/bin/bash
source $HOME/opt/root/bin/thisroot.sh
source $HOME/opt/geant4/bin/geant4.sh
export PYTHIA8DIR=$HOME/opt/pythia8
export HEPMCDIR=$HOME/opt/hepmc
export PATH=$PYTHIA8DIR/bin:$HEPMCDIR/bin:$PATH
export LD_LIBRARY_PATH=$PYTHIA8DIR/lib:$HEPMCDIR/lib:$LD_LIBRARY_PATH
```

Then source ~/.sphenixrc at the top of every work session. Do **not** put this in ~/.bashrc unless you want every terminal to load it.

7. Sanity Checks

After installation, verify each tool works standalone before moving to the next module:

```
which g++ gfortran cmake
```

```
gcc --version
root --version    # after ROOT install
pythia8-config --version  # after Pythia install
geant4-config --version   # after Geant4 install
```

Note: If any 'which' command returns nothing, the tool is not in your PATH — source the correct environment file before running the check.